

Frame-aware CNN-Mamba

Gene-signal prediction and strict locus reconstruction

Drosophila chrX benchmark • UniAnn integration • human transfer gap

CNN-MAMBA-k7

FRAME DILATION 3

PHASE AUXILIARY

STRICT GFFCOMPARE

Architecture ablations, window-size search, auxiliary-loss grids, and end-to-end annotation

Two evaluation regimes answer different questions

Held-out chrX measures generalization; chrX-in-training supports UniAnn assembly

MODEL EVALUATION

Train: 2L, 2R, 3R, 4, Y

Validation: 3L

Test: chrX (never seen in training)



Candidate-level metrics

Exact AUPRC and PR-Sn curves

Donor • acceptor • start • stop

ANNOTATION EVALUATION

chrX is allowed in training

Score every canonical chrX+ candidate

Run UniAnn and strict gffcompare



Gene-model metrics

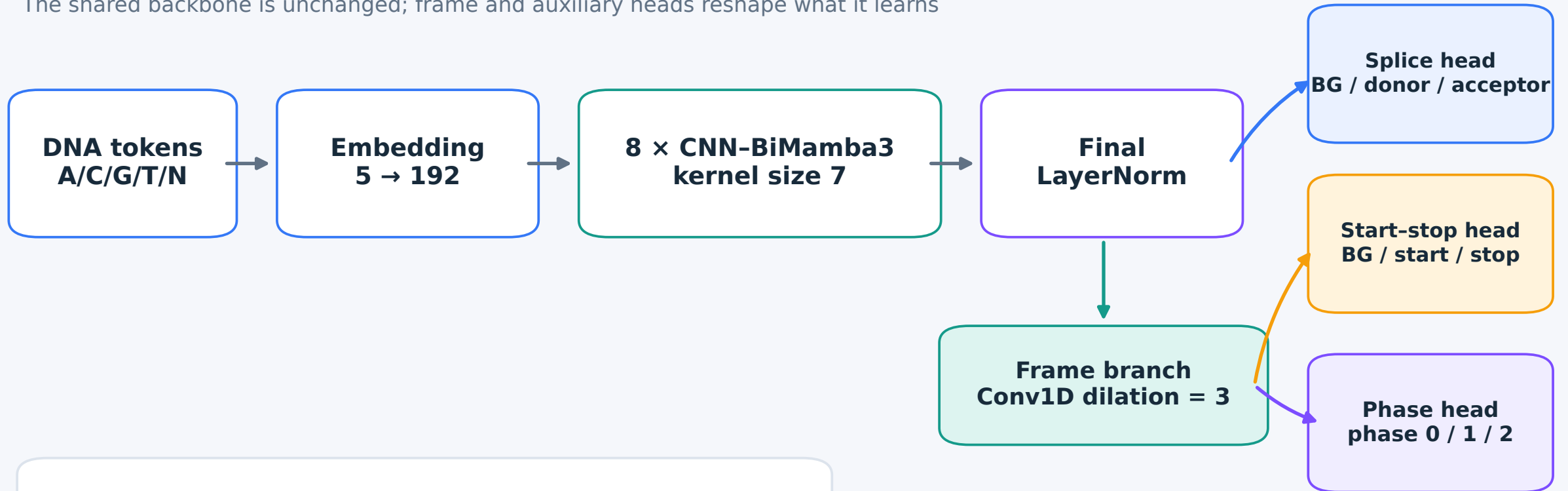
gffread -M → gffcompare --strict-match -e 0

Sn • Pr • F1 at locus level

The two regimes must not be mixed: chrX-in-training scores are not evidence of held-out generalization.

From sequence context to frame-aware multi-task supervision

The shared backbone is unchanged; frame and auxiliary heads reshape what it learns



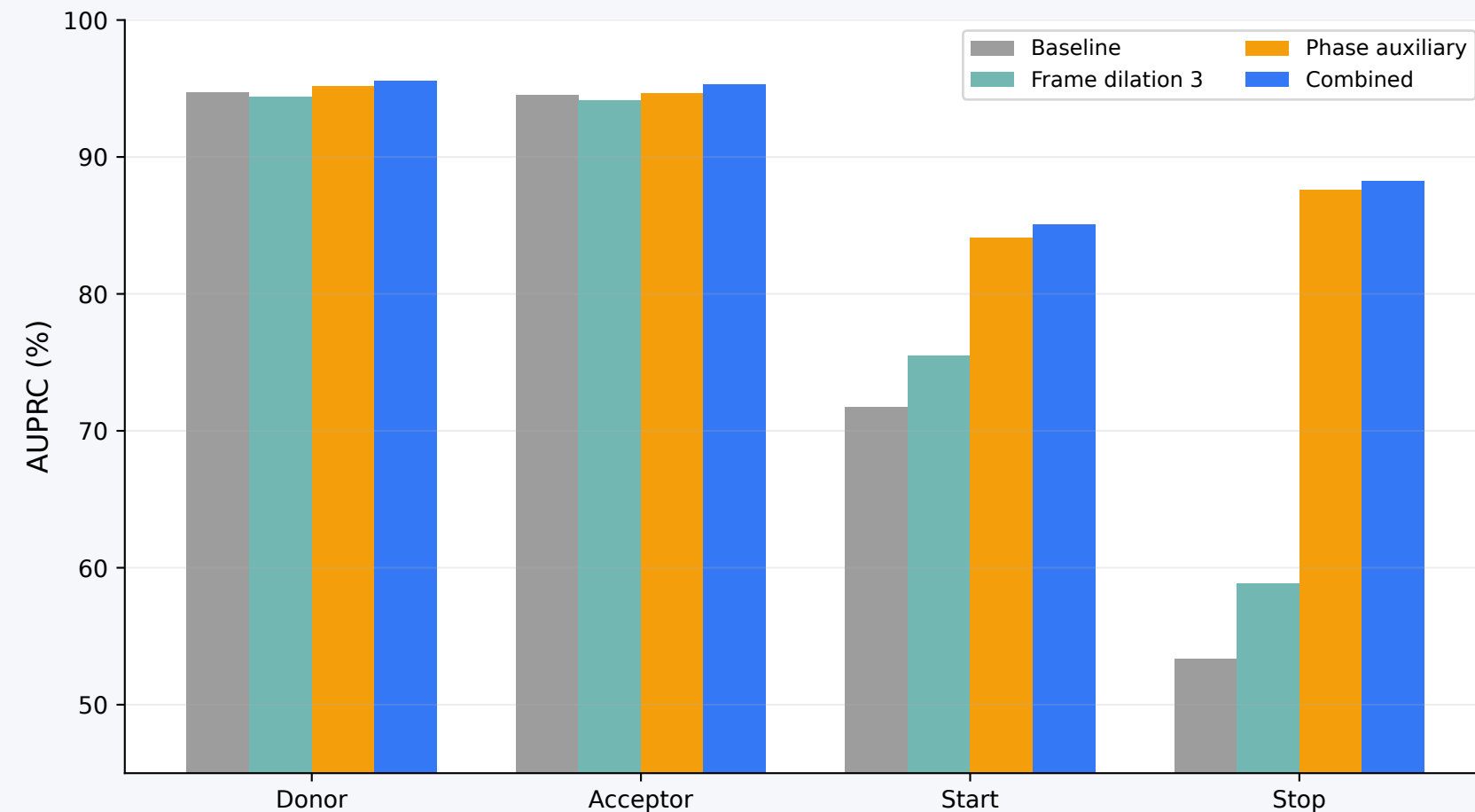
Training objective

$$L = 0.25 L_{\text{splice}} + L_{\text{start/stop}} + 0.10 L_{\text{phase}}$$

Phase labels follow annotated CDS across introns.
The auxiliary head is used during training; UniAnn receives only donor, acceptor, start, and stop candidate scores.

Frame and phase supervision drive the largest gain

Exact positive-strand candidate AUPRC; chrX excluded from training



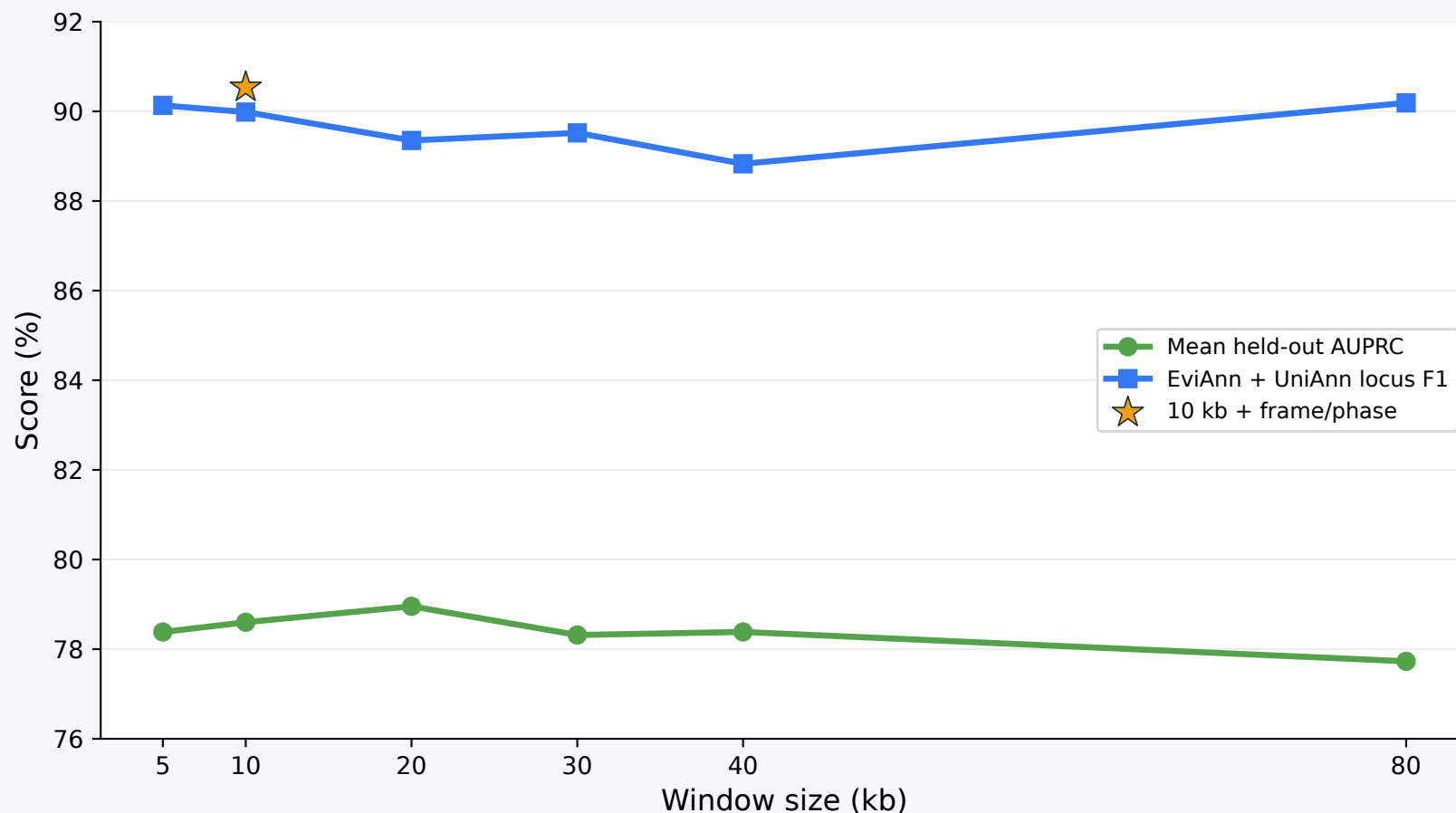
START
71.8 → 85.1
+13.3 points

STOP
53.4 → 88.2
+34.9 points

FOUR-TASK MEAN
78.6 → 91.0
+12.4 points

Longer windows are not the primary performance driver

Same CNN-Mamba-k7 setup; early stopping applied independently



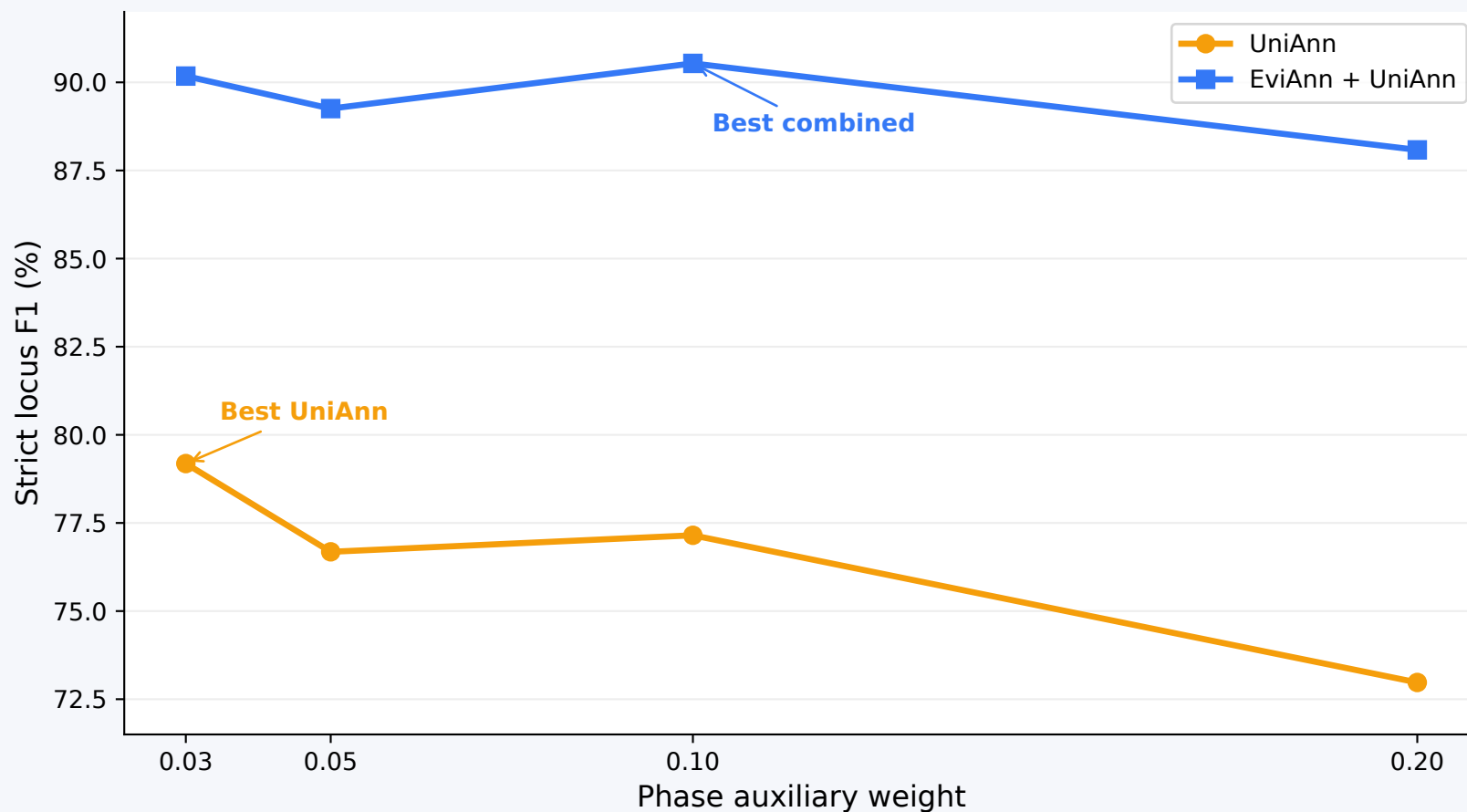
Best raw mean AUPRC
20 kb: 78.96%

Best final locus F1
10 kb + frame/phase: 90.54%

Increasing context alone is not monotonic.
Biological supervision matters more than
window length.

Phase-loss weight controls a recall-precision trade-off

Architecture fixed; only λ_{phase} changes



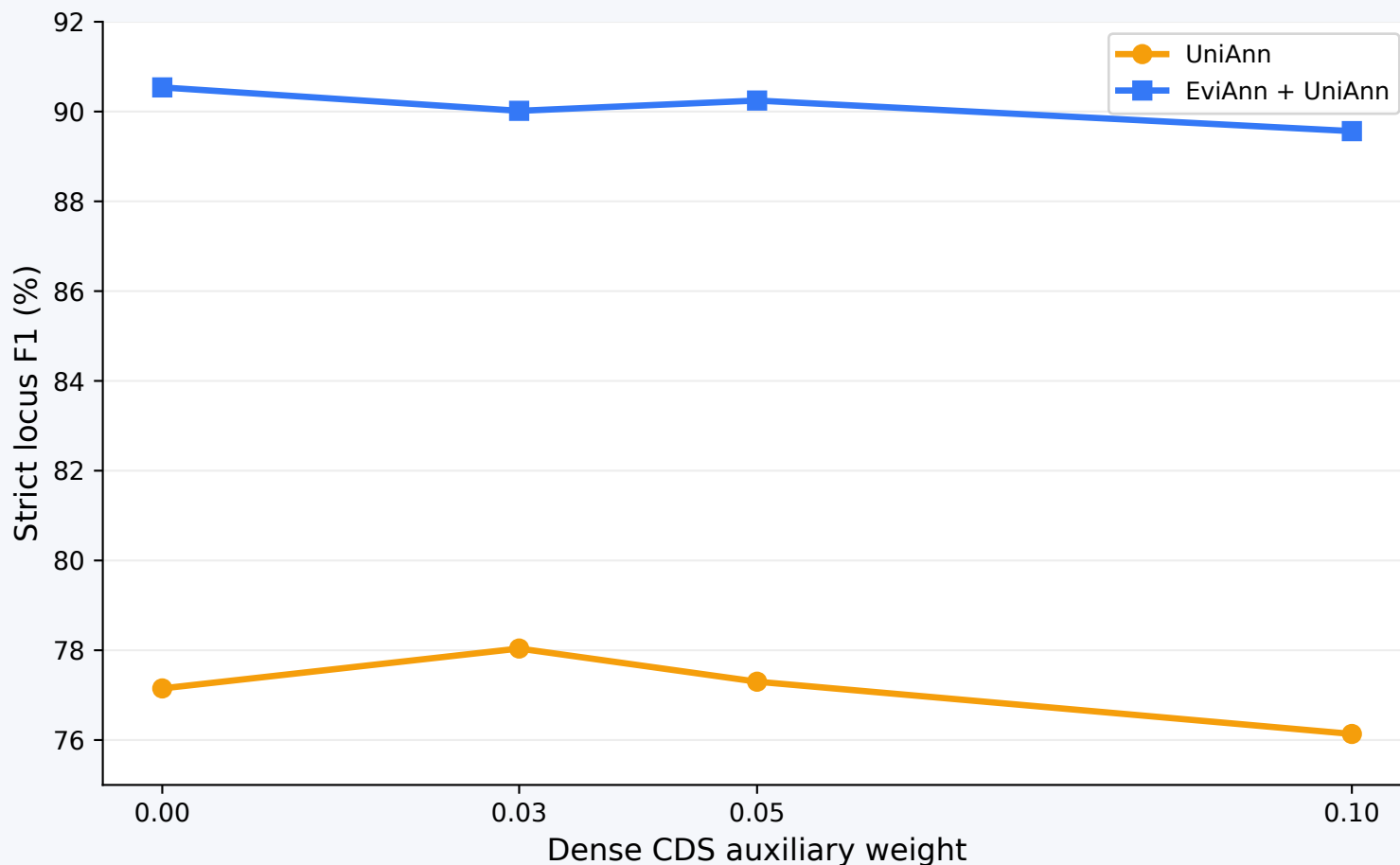
$\lambda_{\text{phase}} = 0.03$
UniAnn Sn / Pr / F1
80.3 / 78.1 / 79.18

$\lambda_{\text{phase}} = 0.10$
EviAnn + UniAnn Sn / Pr / F1
91.6 / 89.5 / 90.54

$\lambda_{\text{phase}} = 0.20$ is consistently worse, indicating over-regularization.

Dense CDS supervision improves recall—but not final precision

The CDS head is highly accurate, yet its signal is largely redundant downstream



CDS prediction itself is easy
AP = 0.986-0.990

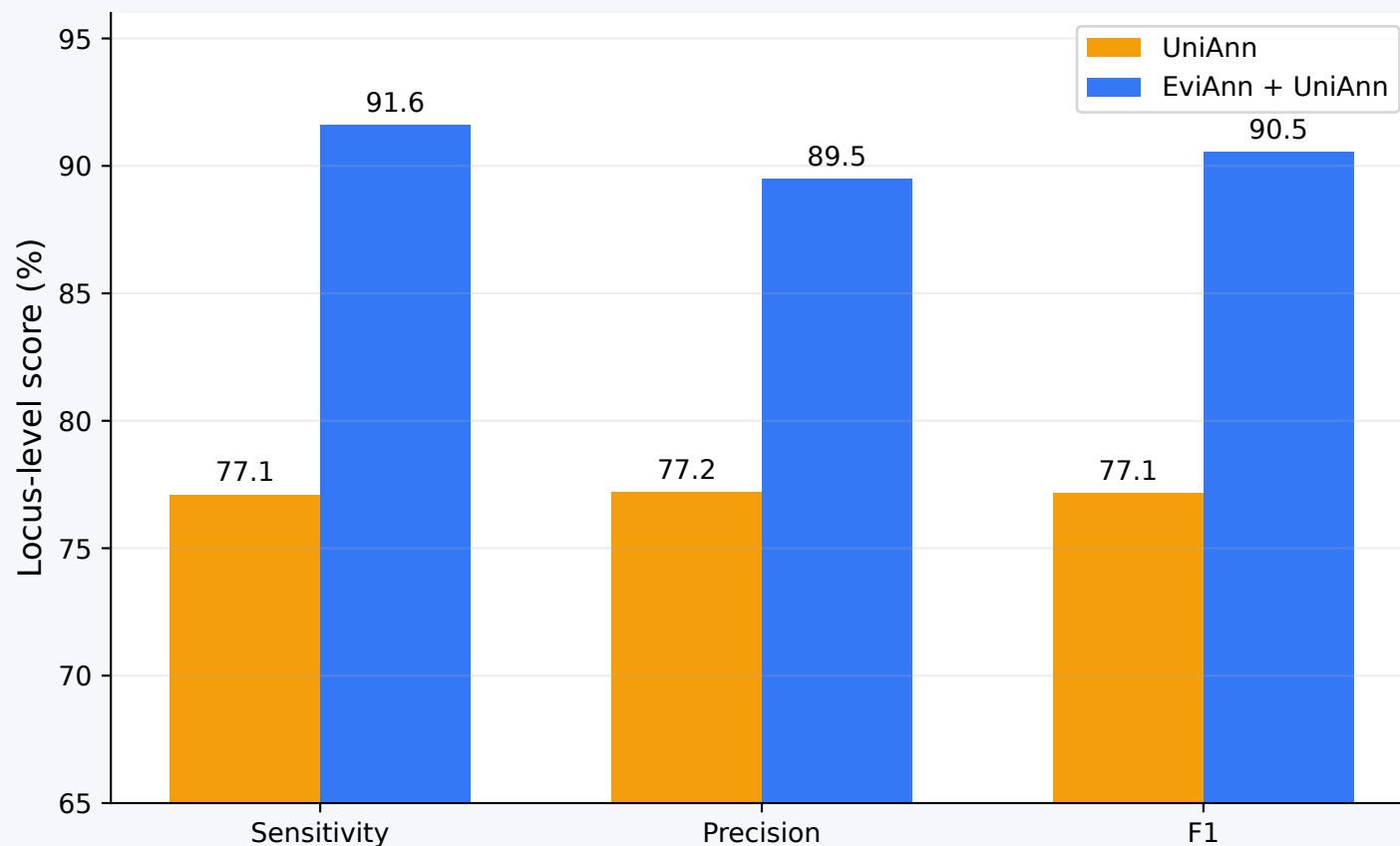
$\lambda_{\text{CDS}} = 0.03$ improves UniAnn
F1: 77.15 $\rightarrow$ 78.04 | Sn: 77.1 $\rightarrow$ 79.0

But final combined F1 falls
90.54 $\rightarrow$ 90.02

Additional true loci mostly overlap EviAnn.
Novel loci increase, and $\lambda_{\text{CDS}} = 0.10$ consumes
~30-40% of the weighted loss.

The frame-aware model improves end-to-end gene reconstruction

10 kb, dilation 3, $\lambda_{\text{phase}} = 0.10$; chrX included for UniAnn scoring



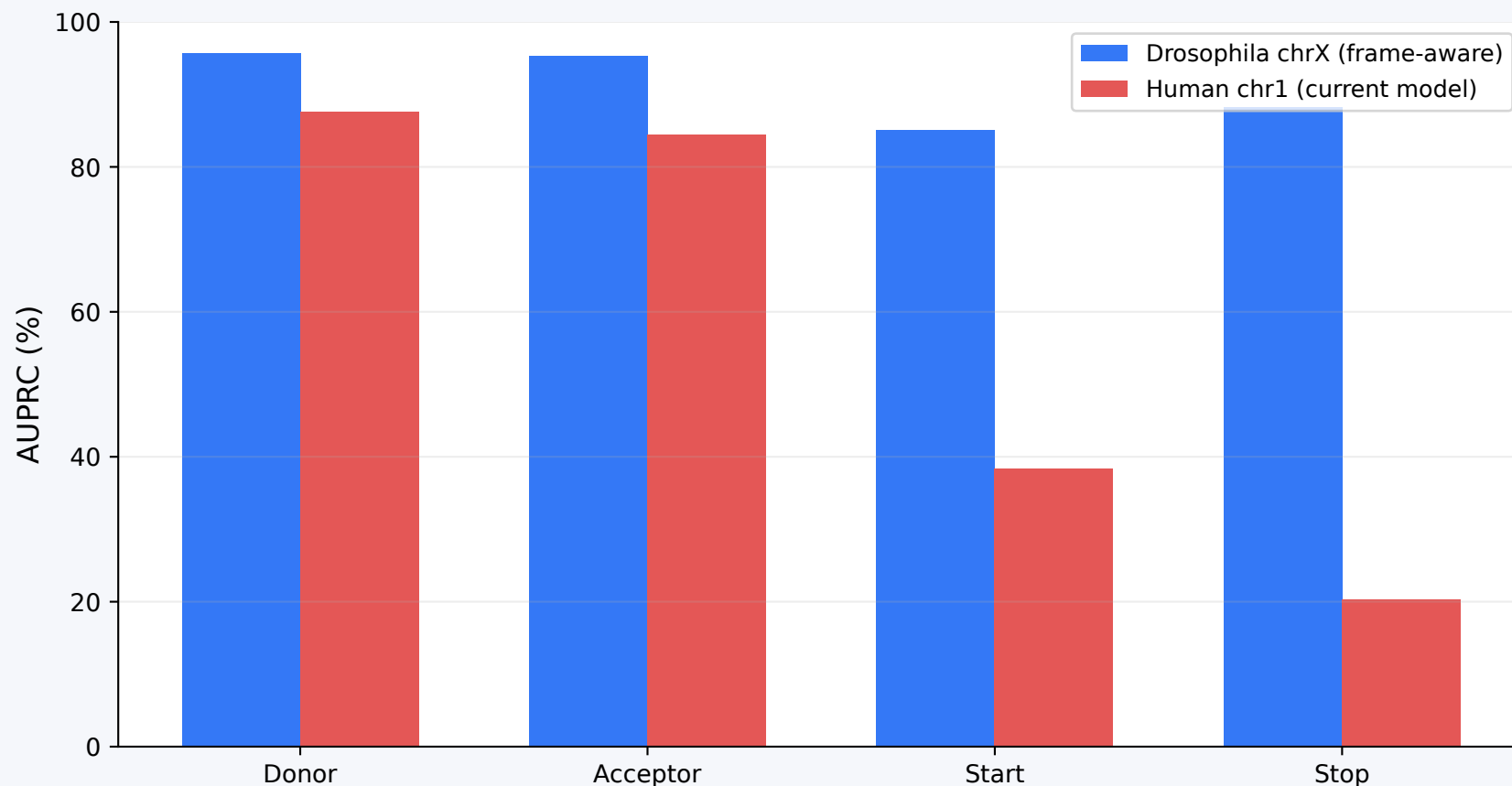
Method	Predicted	Matched	Missed	Novel
UniAnn	1,068	825	41	28
EviAnn + UniAn	1,080	980	20	35

Primary outcome

**EviAnn + UniAnn
strict-locus F1 = 90.54%**

The human transfer gap is concentrated at translation boundaries

Current human chr1 benchmark; candidate-level AUPRC



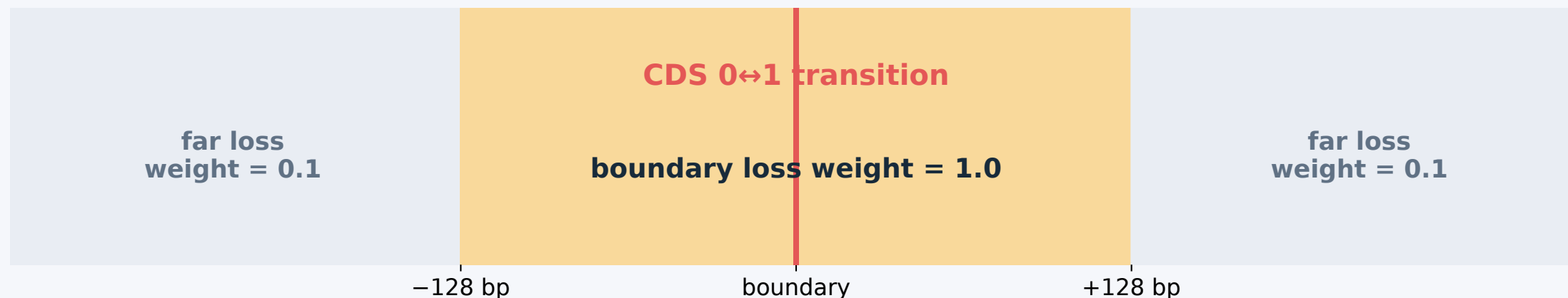
Splice sites remain usable
Donor 87.5% | Acceptor 84.4%

Translation boundaries collapse
Start 38.3% | Stop 20.3%

This motivates explicit coding-frame and boundary supervision. The cross-species bars are diagnostic, not a controlled pair.

Next experiment: concentrate CDS supervision near boundaries

Reduce redundant gradients from easy CDS interiors



Boundary-focused CDS loss

$$L_{\text{CDS}} = \text{mean}(\text{CE}_{\text{boundary}}) + 0.1 \text{ mean}(\text{CE}_{\text{far}})$$

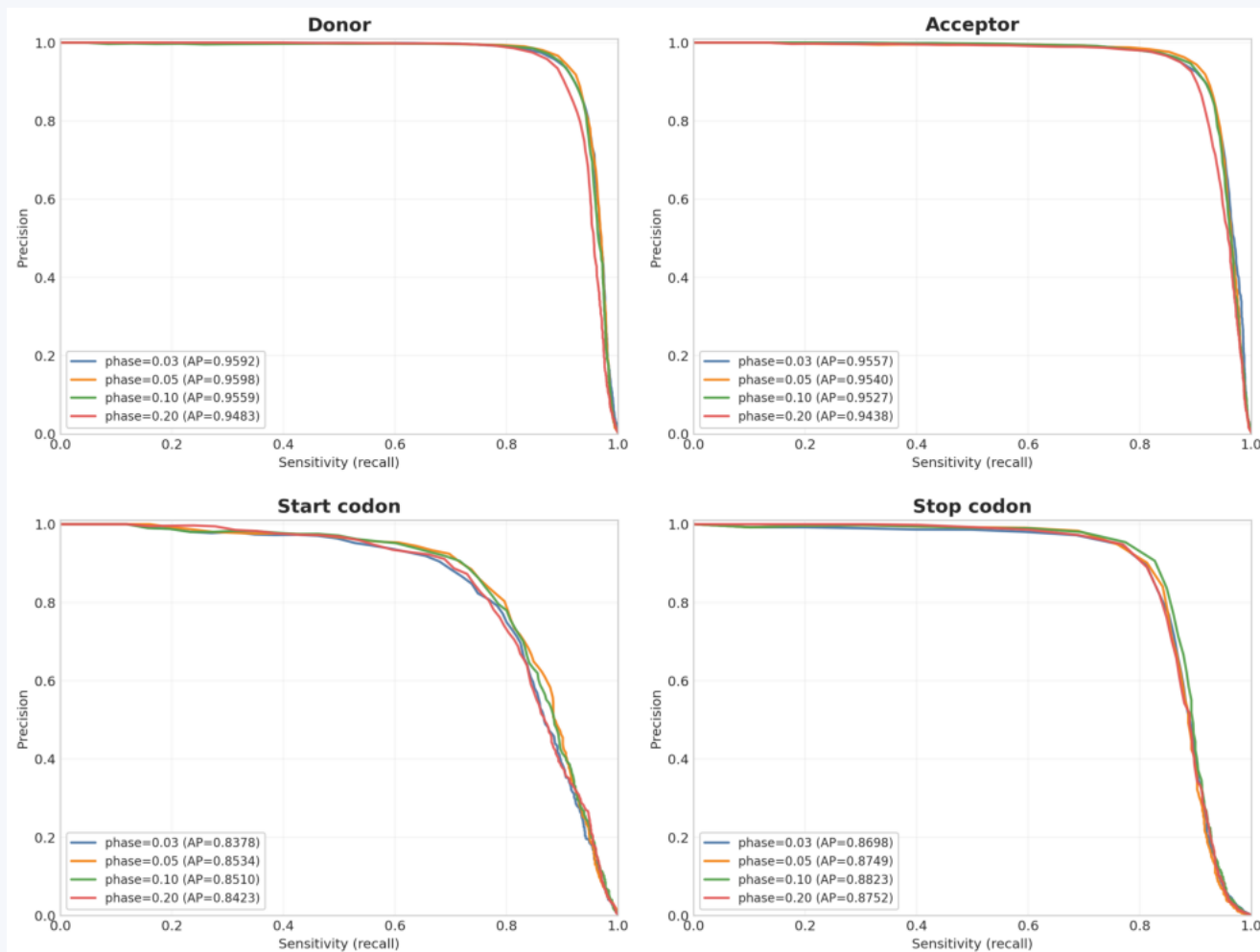
Controlled setting

$$\lambda_{\text{CDS}} = 0.03 \cdot 10 \text{ kb} / 5 \text{ kb stride}$$
$$\text{phase} = 0.10 \cdot \text{dilation } 3 \cdot \text{seed } 42$$

Hypothesis: retain the UniAnn sensitivity gain while reducing novel loci and recovering precision. Results pending.

Candidate-level PR-Sn curves support the held-out conclusion

Phase-loss grid on chrX+; chrX excluded from training



Splice-site curves are nearly saturated

Phase weight separates start/stop behavior

$\lambda_{\text{phase}} = 0.20$ degrades all four tasks

CONCLUSIONS

Take-home messages

What is established, and what remains to be tested

1 Frame awareness works

Phase supervision produces the largest gains, especially for start and stop codons.

2 Context length is secondary

Larger windows do not improve candidate or locus metrics monotonically.

3 Downstream metrics matter

AUPRC gains do not automatically improve strict locus precision after EviAnn merging.

4 Human transfer remains difficult

The largest transfer failure is at translation boundaries, motivating boundary-focused training.

Best completed Drosophila endpoint: EviAnn + UniAnn strict-locus F1 = 90.54%

Ongoing: chromosome-aware boundary-weighted CDS auxiliary training